

KARANDEEP BHARDWAJ

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SUMMARY

Lead Software Engineer and AI systems architect with 7+ years delivering production AI systems and cloud-native platforms that serve enterprise-scale business needs. Specialized in **LLM-based agentic workflows, RAG pipelines, and serverless architecture**, with microservices processing 600k+ daily requests. Expert in **AWS (Bedrock, Lambda, Lex, AgentCore)**, modern full-stack development (Python, TypeScript, React), and architecting **multi-agent AI systems**. Delivered **80% reductions in agent handle time** through LLM-driven automation; experienced in driving technical roadmaps, cross-functional collaboration, and growing high-performing engineering teams.

EXPERIENCE

Lead Software Engineer

Nov 2021 – Present

SunLife Financial

- Architected an AI-powered call routing system using **LangGraph + LLM**, cutting contact center agent handle time by **80%** and processing retirement plan inquiries through intelligent authentication workflows.
- Designed and productionized an **MCP integration** enabling dynamic tool orchestration across multi-agent systems, with a custom middleware layer enforcing 3-tier authentication (identity, single-factor, multi-factor) for enterprise-grade security.
- Spearheading on-premises to cloud migration onto **Amazon Bedrock AgentCore**, deploying agents via AgentCore Gateway and Runtime to achieve fully managed, scalable agentic infrastructure with centralized observability.
- Led migration from legacy systems to **serverless architecture** processing 600k+ daily requests, achieving **99.99% availability** with 30% lower latency.
- Built and deployed an **agentic workflow on Amazon Bedrock** with DynamoDB-backed state persistence, routing 15+ retirement plan categories through context-aware conversation flows.
- Optimized **AWS End User Messaging (SNS)** and multi-Lambda orchestration via Amazon Connect (Lex, ANI lookup, OTP verification), reducing authentication resolution time by **65%**.
- Led and mentored a team of 5+ engineers on **LLM-based system design**, agentic architecture patterns, and serverless best practices – owning the technical roadmap and presenting quarterly AI strategy to senior leadership.

Full Stack Engineer

Jan 2021 – Nov 2021

Tecsys Inc.

- Led a full monolith-to-microservices re-architecture using **event-driven architecture** (Kafka + SQS), scaling the system to handle **50x concurrent requests** and increasing throughput by **85%**.
- Led front-end modernization using **Storybook and atomic design**, building a reusable component library adopted across 3+ product teams and boosting development velocity by **25%**.
- Engineered an end-to-end customs compliance microservice using **AWS SAM, Lambda, SQS, and S3** with SFTP integration, eliminating a third-party vendor dependency and delivering **\$900K in validated annual cost savings**.
- Established an org-wide testing strategy using **TDD (Jest)** and k6 load testing, achieving zero P0 incidents during peak traffic windows and reducing post-release defect rate across the engineering team.
- Hardened integration security by implementing **OAuth 2.0** across 10+ carrier integrations, managing token lifecycle (issuance, refresh, revocation) across distributed microservices.

Full Stack Engineer

Jan 2020 – Jan 2021

BIDGALA (concurrent with Master of Computer Science, 2018–2020)

- Architected and shipped a full-stack e-commerce social platform using **Python (Django)** and React, with RESTful APIs deployed on **AWS (EC2, RDS, S3)**, supporting early-stage product growth.
- Defined technical architecture and mentored 3 junior developers, establishing code review standards and Git workflows that reduced integration conflicts by **40%**.

Software Developer

Jan 2017 – Aug 2018

Unisys Pvt. Ltd.

- Developed enterprise **Java** applications for Fortune 500 clients in cross-functional Agile squads, contributing across the full SDLC from requirements to deployment and establishing coding standards adopted team-wide.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, Golang, SQL (Postgres), Ruby on Rails

Frameworks: React, Next.js, Vue, Node.js, Express, Django, Kafka, GraphQL, Tailwind

AI/ML: LangGraph, LangChain, Amazon Bedrock, AgentCore, RAG Pipelines, Vector Databases (pgvector, Pinecone), MCP, Prompt Engineering, AI Observability

Cloud & DevOps: AWS (Lambda, Bedrock, Lex, DynamoDB, S3, SNS, SQS), Docker, Kubernetes, Terraform, Jenkins, CI/CD, GitOps, OpenShift

Architecture: Microservices, Serverless, Event-Driven, Multi-Agent Systems, Agentic Workflows, System Design

Testing: Jest, k6, TDD, Test Automation

EDUCATION & CERTIFICATIONS

Master of Computer Science, 2018 – 2020

Concordia University, Montreal, Canada

Bachelor of Computer Science, 2013 – 2017

Chandigarh Engineering College, Chandigarh, India

AWS Solutions Architect – Associate (2022)